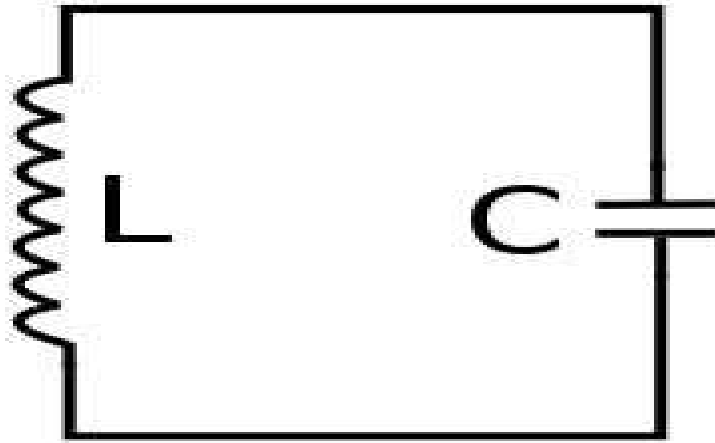


LC Circuits

- A type of circuit that is well-known from classical circuit theory is the *LC circuit*, in which an inductor and a capacitor cause oscillations in the flux of a circuit loop:



- The energy function for this circuit can be written

$$H = \frac{Q^2}{2C} + \frac{\Phi^2}{2L}, \quad \omega = \frac{1}{\sqrt{LC}}.$$

- Here, Q is the charge on the capacitor, Φ is the magnetic flux in the inductor, and C and L are the capacitance and inductance.
- We can make circuits smaller and smaller, so that Q and Φ become noncommuting quantum observables:

$$[\hat{\Phi}, \hat{Q}] = i\hbar.$$

But any quantum effects are masked by resistance, which causes decoherence.

- We can get rid of this decoherence by going to very low temperatures where our circuit is *superconducting*. The electrons become bound into *Cooper pairs*, and all resistance vanishes. The phase is well-defined and the system is a quantum harmonic oscillator.

Josephson Junctions

- A superconducting LC circuit would not work as a q-bit, because (like any harmonic oscillator) it has infinitely many evenly-spaced energy levels; resonant driving does not let us single out two levels to use as $|0\rangle$ and $|1\rangle$, nor perform gates on this subspace alone.
- We can fix this problem by adding a *Josephson junction* to the circuit. This is a thin insulating barrier between two ends of superconductor. Classically this would act as either an open circuit, or at best a capacitor; but in the quantum regime, Cooper pairs can tunnel across the gap, so a current can flow.
- The effect of including a Josephson junction is to add a nonlinear term to the harmonic oscillator potential.

- The potential with the added Josephson junction takes this form:

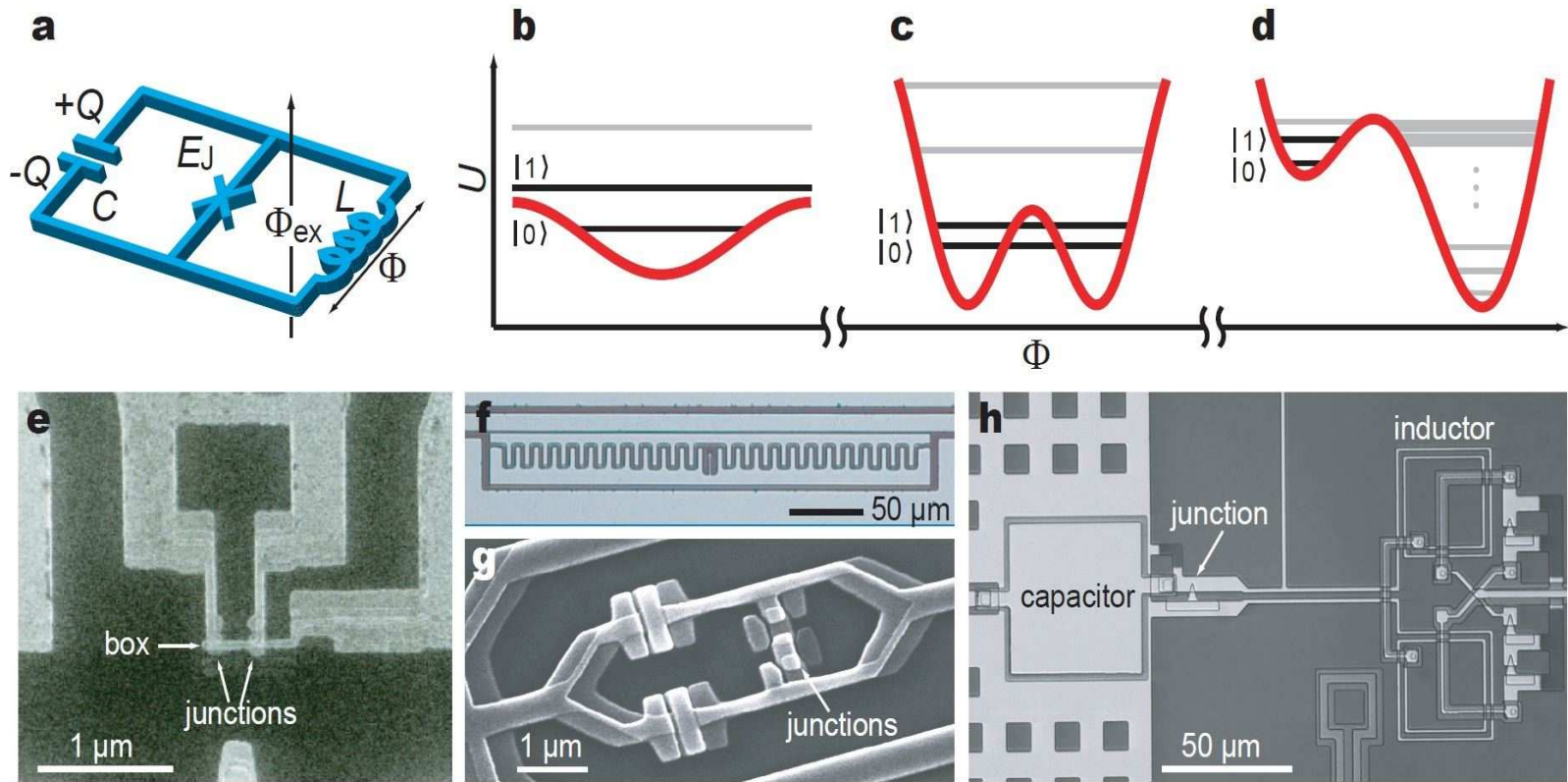
$$U(\Phi) = E_J \left[1 - \cos \left(2\pi \frac{\Phi_{\text{ex}} - \Phi}{\Phi_0} \right) \right] + \frac{\Phi^2}{2L'},$$

where $\Phi_0 = h/2e$ is the *quantum of flux*, Φ_{ex} is the external bias flux, and E_J is the Josephson energy, which is proportional to the critical current through the junction.

- The behavior of the circuit is determined by the applied bias flux Φ_{ex} and the ratio E_J/E_C where E_C is the capacitor charging energy $E_C = e^2/2C$. This leads to three different designs for superconducting q-bits.

Superconducting Q-bits

We can put these elements together to get different kinds of q-bits, characterized by the form of the potential.



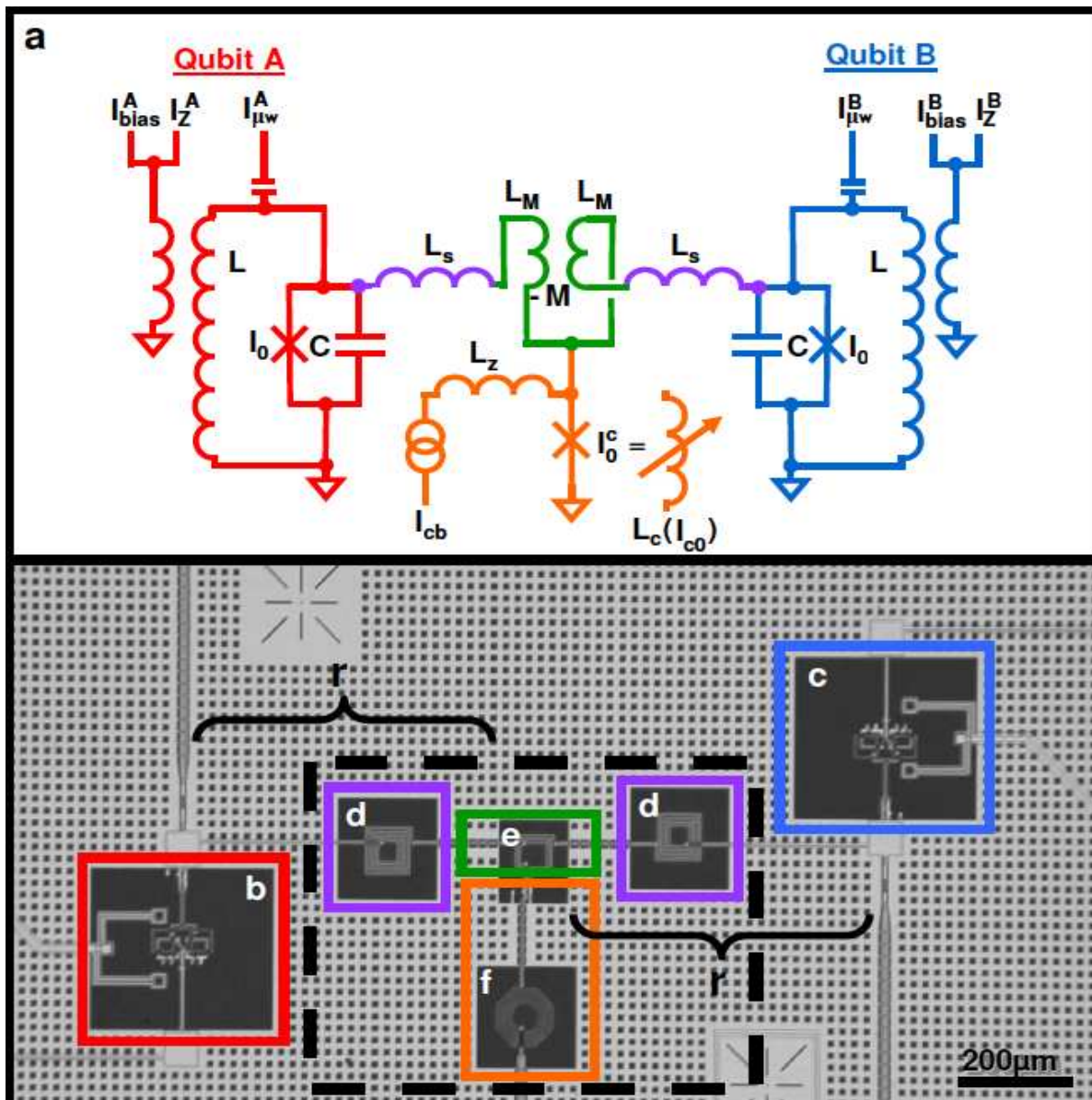
- The three q-bit types are the *charge* q-bit, the *flux* q-bit, and the *phase* q-bit.
- *Charge q-bit* (Cooper-pair box): omits the inductance L and uses the two lowest levels of the cosine potential for $|0\rangle$ and $|1\rangle$. These represent the absence and presence of a single extra Cooper pair on the “island” of superconducting metal.
- *Flux Or persistent-current q-bit*: uses a double-well potential representing current circling clockwise or counterclockwise.
- *Phase q-bit*: uses bias to produce unequal wells. The lowest two levels of the *higher* well are $|0\rangle$ and $|1\rangle$; the lower well is used to make measurements.

Single-bit gates

- The level separations in superconducting q-bits are typically in the range 5-10 GHz (microwaves). Single-bit quantum gates can be done using *resonant driving*, just as for the ion trap, but at much lower frequencies.
- A gate is performed by applying a pulse of RF, tuned to the precise resonance frequency between (e.g.) $|0\rangle$ and $|1\rangle$. This can produce an X or Y gate. A Z gate can be done by using the energy splitting between $|0\rangle$ and $|1\rangle$ to produce a relative phase.
- The pulse can be produced by an external signal generator, and carried to the q-bit by wires. RF engineering can be done with great precision; these pulses have very exact timing and frequency control.

Two-bit gates

- Superconducting q-bits are (relatively) macroscopic objects—a typical q-bit is μms or 10s of μms in size. Therefore it is straightforward to couple nearby q-bits either capacitively or inductively. A disadvantage of this scheme, however, is that only neighboring q-bits can be coupled, and the coupling is always turned on. (This is not necessarily a fatal problem, as we will see in NMR.)
- A more flexible scheme uses a *tuneable coupler*. This is an inductance that can be controlled by externally applied fields. It can also couple q-bits that are not directly adjacent. This has been demonstrated experimentally for phase q-bits.



- It is also possible to couple two q-bits indirectly, by coupling each of them to the same *transmission line resonator*. This acts like a bus, similar to the vibrational mode in an ion trap quantum computer.
- By resonant driving, one can conditionally excite a microwave photon in the resonator from one q-bit, absorb and re-emit it from another, and reabsorb it at the first, to produce an effective gate for 2 q-bits.
- One can also couple two q-bits by tuning them close to resonance, but far enough that no actual photon is emitted or absorbed. This is slower than the first scheme, but less sensitive to photon loss.
- Transmission line resonators can couple q-bits that are physically far apart—it has been demonstrated at distances of millimeters.

Measurement

- Measurement of superconducting q-bits is one of the most difficult problems, because it is difficult to do without allowing in outside noise, causing decoherence.
- Charge q-bits can be measured by charge detectors.
- Flux q-bits have been measured in different ways, but generally by inductive coupling to a nearby microscopic device, such as a SQUID.
- Phase q-bits can be measured destructively by lowering the energy barrier to induce tunneling into the deeper well. This produces a detectable current, but destroys the state. Nondemolition measurement has been demonstrated more recently by coupling to a transmission line resonator to produce a state-dependent phase shift.

Decoherence

- Because superconducting q-bits are macroscopic objects, they are subject to many sources of noise. Early q-bits had very short decoherence times—on the order of 1 ns. Improved design and fabrication have greatly improved this, bringing lifetimes up to $\sim 1\mu\text{s}$.
- The exact sources of decoherence are not completely known. Experimentalists believe that the q-bits couple to microscopic degrees of freedom that arise from defects in the bulk substrate, in the insulating layer, and in imperfections in the circuit construction.
- Measurement can also open up new sources of noise—for instance, thermal noise from an external amplifier can leak down the transmission line.

State of the Art

- Superconducting q-bits have made the most impressive progress in the last few years, raising their decoherence times by orders of magnitude and demonstrating coupling and entanglement between up to four q-bits. There seems to be no barrier to scaling this up to many more q-bits, though decoherence remains an issue.
- Superconducting q-bits are now challenging ion traps as the most promising technology for scalable quantum computation. While they started much further behind, they have certain advantages: solid state fabrication is constantly improving, and their intrinsic “clock speeds” are much faster.

The D-Wave Chip



- A private company, D-Wave, has built a chip with 128 qubits! This is not a general-purpose quantum computer, however, but rather an *adiabatic* quantum computer, designed to solve one particular class of problems: to find the *ground state of a generalized Ising model* on a particular class of *partially-connected graphs*. Many classical optimizations can be reduced to this problem. The first chip outside of the company is at USC's Information Sciences Institute in Marina Del Rey.

NMR Quantum Computing

- In NMR quantum computing, the computer is a single molecule of a chemical compound. The q-bits are the nuclear spins of the atoms in the molecule (or at least those which are spin-1/2). Quite a few isotopes have spin-1/2 nuclei, notably hydrogen and carbon-13.
- A powerful magnetic field is introduced in the Z direction, resulting in a separation of energy between spin-up and spin-down. This separation is important for practical purposes.
- Unlike other models we have considered, however, NMRQC is done with *many* (e.g., 10^{24}) molecules in the liquid state, all performing the same calculation in parallel. NMR computation is done with a bulk sample at room temperature.

- Perhaps surprisingly, despite being immersed in a bulk at room temperature, the nuclear spins can be treated as isolated to a very good approximation. This is because (a) the nuclear magnetic moments are very weak, which means they evolve on a very slow timescale, (b) the nuclei are surrounded by rapidly moving clouds of electrons which screen out other nearby charges, and (c) the molecules tumble rapidly, so that surrounding potentials are averaged out on a timescale short compared to that of the spin evolution.
- Because of these effects, it is possible to treat the nuclear spins using the Schrödinger equation; a fact which chemists have been exploiting for years. To analyze them we need to know their Hamiltonian to a good approximation.

The Spin Hamiltonian

- The presence of the magnetic field together with the (much weaker) Hamiltonian due to the dipole-dipole interactions of the spins produces a Hamiltonian of the form

$$\hat{H} = \sum_j E_j \hat{Z}_j + \sum_{i,j} \alpha_{ij} \hat{\vec{\sigma}}_i \cdot \hat{\vec{\sigma}}_j \equiv \hat{H}_B + \hat{H}_{SS}.$$

- The E_j will be similar (but not identical) for all nuclei of the same chemical species, and the interaction strength α_{ij} between nuclei i and j will depend on their chemical species and distance from each other in the molecule. These parameters are measured experimentally.
- Because $E_j \gg \alpha_{ij}$, the evolution is dominated by the magnetic field, which produces spin precessions on a timescale fast compared to the spin-spin interactions.

RF pulses and 1-bit gates

- Just as in the ion trap, single-spin transitions can be produced by *resonant driving*. In this case, they are transitions between, e.g., Z -up and Z -down states for spin j , and have resonant energy $2E_j$. As long as E_j is sufficiently different for different spins, one can tune the resonance so that it affects only a single spin. (It is also possible to tune pulses that affect all spins of a single chemical species.) These pulses can (in principle) produce any single q-bit rotation: $R_{x,y,z}(\theta)$. This makes any one-bit gate possible.
- Unlike the ion trap, however, the energy differences $2E_j$ are very small in NMR; so the resonant pulses are not at optical frequencies, but at *radio* frequencies. These RF pulses are sufficiently brief that we will approximate them as instantaneous unitaries.

The interaction picture

- We do two-bit gates using the natural spin-spin interaction of the molecule. We see how this works by transforming to a *rotating* picture: the *interaction picture*.
- Let $|\psi(t)\rangle$ be the solution to the Schrödinger equation with the full Hamiltonian $\hat{H}$. Define the *rotating state* $|\tilde{\psi}(t)\rangle$

$$|\tilde{\psi}(t)\rangle = \exp(i\hat{H}_B t/\hbar)|\psi(t)\rangle,$$

which obeys the new Schrödinger equation

$$\frac{d|\tilde{\psi}(t)\rangle}{dt} \approx -(i/\hbar)\hat{H}'_{SS}|\tilde{\psi}(t)\rangle = -(i/\hbar)\sum_{ij}\alpha_{ij}\hat{Z}_i\hat{Z}_j|\tilde{\psi}(t)\rangle,$$

with only ZZ terms and no explicit time-dependence.

Refocusing and 2-bit gates

- Hamiltonian terms like $\hat{Z}_i \hat{Z}_j$ can produce an entangling two-bit gate. Together with one-bit unitaries these are universal for quantum computation:

$$\exp(-i\pi \hat{Z}_i \hat{Z}_j / 4) = (\hat{I} - i\hat{Z}_i \hat{Z}_j) / \sqrt{2}.$$

The problem is that we can't turn these terms off! The spins interact continuously, outside of our control.

- There is a technique, however, whereby all but a single term in the Hamiltonian can be made to effectively cancel out. This is called *refocusing*. To illustrate this, suppose we have a Hamiltonian with only two terms:

$$\hat{H} = \alpha_{12} \hat{Z}_1 \hat{Z}_2 + \alpha_{23} \hat{Z}_2 \hat{Z}_3.$$

After evolving for a time 2τ , the unitary transformation would be $\hat{U} = \exp(-2i\tau(\alpha_{12}\hat{Z}_1\hat{Z}_2 + \alpha_{23}\hat{Z}_2\hat{Z}_3))$. If we applied an $\hat{X}$ gate to spin 3 at time $t = 0$ and again at $t = \tau$ the new evolution would be

$$\begin{aligned}\hat{U}' &= \exp(-i\tau(\alpha_{23}\hat{Z}_2\hat{Z}_3))\hat{X}_3 \exp(-i\tau\alpha_{23}\hat{Z}_2\hat{Z}_3)\hat{X}_3 \\ &\quad \times \exp(-2i\tau\alpha_{12}\hat{Z}_1\hat{Z}_2).\end{aligned}$$

Since $\hat{X}_3\hat{Z}_3\hat{X}_3 = -\hat{Z}_3$, we know that $\hat{X}_3 \exp(-i\tau\alpha_{23}\hat{Z}_2\hat{Z}_3)\hat{X}_3 = \exp(+i\tau\alpha_{23}\hat{Z}_2\hat{Z}_3)$. So the terms with $\hat{Z}_2\hat{Z}_3$ cancel, leaving

$$\hat{U}' = \exp(-2i\tau\alpha_{12}\hat{Z}_1\hat{Z}_2).$$

We have effectively “turned off” the term we didn’t want.

Thermal and Effective Pure States

- Because the molecules are at room temperature, the state of the spins will not be pure; they will be in a particular mixed state called a *thermal state*.

$$\rho_T = \frac{1}{Z(T)} \exp(-\hat{H}/k_B T),$$

where $\hat{H}$ is the Hamiltonian, T is the temperature, and k_B is Boltzmann's constant. $Z(T) = \text{Tr}\{\exp(-\hat{H}/k_B T)\}$ is a normalization factor called the *partition function*.

- Generally speaking, it is impossible to do quantum computation in a thermal state, unless the temperature is very low. Nevertheless, there is a trick that allows interesting computations to be done. This is the preparation of an *effective pure state*.

- An effective pure state has the form $\rho = (1 - \epsilon)\hat{I}/2^n + \epsilon|\psi\rangle\langle\psi|$, where $\epsilon \ll 1$. A state of this type undergoes unitary transformations

$$\rho \rightarrow \hat{U}\rho\hat{U}^\dagger = (1 - \epsilon)\hat{I}/2^n + \epsilon\hat{U}|\psi\rangle\langle\psi|\hat{U}^\dagger.$$

- We prepare the effective pure state from an initial thermal state by applying different unitaries to different molecules in the sample, and average the signals from all of them. The average of these unitaries will *not* be unitary, and if we choose them carefully they can produce an effective pure state. (In practice, this might be done by imposing a magnetic field that is different at different points in the sample.)
- Generally we choose $|\psi\rangle\langle\psi| = |0 \cdots 0\rangle\langle 0 \cdots 0|$ at $t = 0$.

Readout

- There is another reason to use a bulk sample: we have no way of measuring the state of the nuclear spins for a single molecule. They are too well isolated from the outside world, and the RF signals a single molecule produces are too weak to be detected.
- Instead, the quantum computation is performed on the bulk sample, and then the sample is allowed to relax back to the thermal state. As the spins relax, they emit RF signals at the frequencies $2E_j/\hbar$.
- The contribution to the signal from the identity cancels out. In other words, only the fraction of systems in state $|\psi\rangle$ produce a net signal. By running the computation a number of times and applying different unitary transformations at the end, it is possible to completely reconstruct the final state $|\psi_f\rangle$.

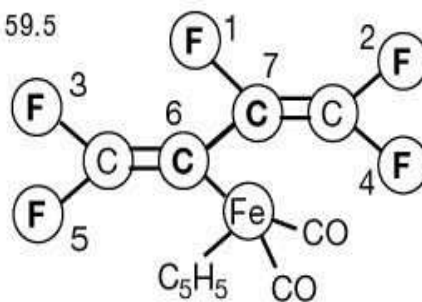
Decoherence

- There are two major decohering effects in NMR. The first is *thermal relaxation*: the spins can slowly gain or lose energy from the surrounding electrons or other molecules, returning to a thermal state. The second is *dephasing*, in which $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ undergoes $|\psi\rangle\langle\psi| \rightarrow |\alpha|^2|0\rangle\langle 0| + |\beta|^2|1\rangle\langle 1|$.
- These two processes have characteristic timescales T_1 and T_2 , respectively. T_1 can be as long as hours or days (though seconds is more typical), and T_2 can be as long as tenths of a second. This gives enough time to perform hundreds or thousands of two-bit gates before the state is destroyed by decoherence.

Experimental set-up

- Because liquid-state NMR was developed by chemists decades ago, one can do NMRQC experiments using off-the-shelf equipment.
- Because of this well-advanced equipment and set of techniques, NMR quantum computing made spectacularly rapid early progress compared to other techniques. It is still the case that many quantum protocols and algorithms have only been demonstrated in NMRQC, including Shor's factoring algorithm.
- As we saw in the homework, 15 is the smallest number for which the factoring problem doesn't reduce to a simplified case. In 2001 the first experimental demonstration of factoring was made at IBM, using a seven-bit NMR quantum computer.

i	$\omega_i/2\pi$	$T_{1,i}$	$T_{2,i}$	J_{7i}	J_{6i}	J_{5i}	J_{4i}	J_{3i}	J_{2i}
1	-22052.0	5.0	1.3	-221.0	37.7	6.6	-114.3	14.5	25.16
2	489.5	13.7	1.8	18.6	-3.9	2.5	79.9	3.9	
3	25088.3	3.0	2.5	1.0	-13.5	41.6	12.9		
4	-4918.7	10.0	1.7	54.1	-5.7	2.1			
5	15186.6	2.8	1.8	19.4	59.5				
6	-4519.1	45.4	2.0	68.9					
7	4244.3	31.6	2.0						



The experiment used a simplified version of Shor's algorithm that needed only a relatively small number of RF pulses to carry out. This is still the only practical implementation of Shor's algorithm. (The factors: 3 and 5.)

Scaleability

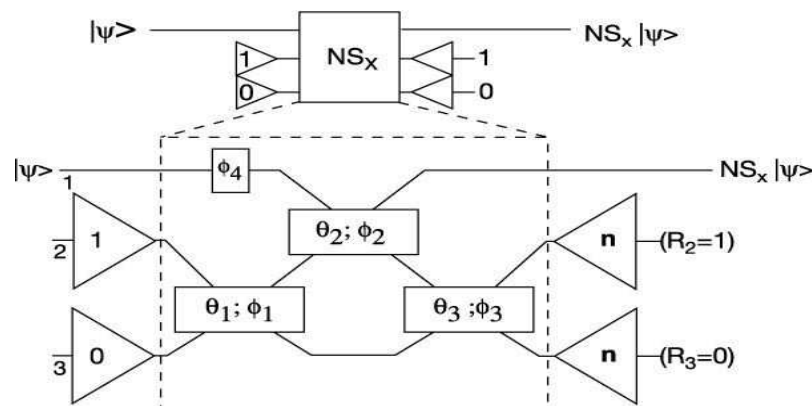
- In spite of this rapid experimental progress, however, as an approach to building large-scale QIP NMR is essentially a dead end.
- In preparing an effective pure state, the largest that ϵ can be is roughly the fraction of systems that are in the correct state $|0 \cdots 0\rangle$ just by chance at the start of the computation. For an n -spin molecule, $\epsilon \sim 2^{-n}$. Since the strength of the signal is proportional to ϵ , as n becomes large the signal goes to zero.
- While the most impressive experiments in QIP to date have been done using NMR, in the long run general purpose quantum computers will have to use different systems. The failure of liquid-state NMR to satisfy the DiVincenzo criteria is a flaw that prevents true scaleability.

Other systems

A number of other systems are being pursued actively as possible technologies for quantum computation. Here are a few of the most promising.

- *Solid-state NMR.* There are various schemes, using the spins of nuclei in crystals; or using the nuclei of phosphorus atoms embedded in silicon, coupled via electron spins.
- *Quantum dots.* These are tiny devices that serve as “wells” to hold electrons. The spin of an electron can serve as the q-bit, or the location of the charge in one of two dots. Like superconducting q-bits, this work draws on the large amount of technological expertise at building ever-smaller solid-state devices.

- *Linear optical QC.* It is possible for photons to serve as q-bits (as we saw in looking at quantum cryptography), and they have very low rates of decoherence, but it is hard to make them interact; this makes it difficult to build two-bit gates. It is possible to build *probabilistic* two-bit gates, however, by passing photons through interferometers and measuring some of the outputs.



It is still too early yet to know which technology may eventually win out.